# DISCOUNTED CASH FLOW VALUATION

- For this valuation I chose Freeport-McMoRan, the 3rd largest copper miner globally.
- I estimated a Value/share of $86 compared to a current market price of $72 suggesting an approximate 15% undervaluation.
- The range of outcomes is large; simulating the copper price inline with a historic mean and standard deviation of 5% and 14% per annum respectively generates a wide range of 10-year Free Cash Flow paths, yet positively skewed.

FREEPORT-MCMORAN

| USD millions | BASE YEAR | YEAR 1 | YEAR 2 | YEAR 3 | YEAR 4 | YEAR 5 | YEAR 6 | YEAR 7 | YEAR 8 | YEAR 9 | YEAR 10 | TERMINAL YEAR |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 | |
| **DISCOUNTED CASH FLOW** | **BASE YEAR** | **YEAR 1** | **YEAR 2** | **YEAR 3** | **YEAR 4** | **YEAR 5** | **YEAR 6** | **YEAR 7** | **YEAR 8** | **YEAR 9** | **YEAR 10** | **TERMINAL YEAR** |
| Net Income | 4,152 | 4,384 | 4,844 | 5,482 | 6,202 | 7,013 | 7,924 | 8,946 | 10,090 | 11,370 | 12,798 | 14,391 |
| D&A | 2,244 | 2,533 | 2,631 | 2,763 | 2,901 | 3,046 | 3,198 | 3,358 | 3,526 | 3,702 | 3,887 | 4,081 |
| Income Tax expense | 2,221 | 2,361 | 2,608 | 2,952 | 3,340 | 3,776 | 4,267 | 4,817 | 5,433 | 6,122 | 6,891 | 7,749 |
| Net interest | 146 | 146 | 237 | 238 | 221 | 185 | 130 | 53 | - 49 | - 176 | - 332 | - 519 |
| **EBITDA** | **8,763** | **9,424** | **10,321** | **11,434** | **12,663** | **14,020** | **15,519** | **17,174** | **19,001** | **21,018** | **23,245** | **25,702** |
| less: | | | | | | | | | | | | |
| Capital expenditures | - 4,494 | - 4,500 | - 4,500 | - 4,500 | - 4,500 | - 4,500 | - 4,500 | - 4,500 | - 4,500 | - 4,500 | - 4,500 | - 4,500 |
| **Free Cash Flow from Operations** | **4,269** | **4,924** | **5,821** | **6,934** | **8,163** | **9,520** | **11,019** | **12,674** | **14,501** | **16,518** | **18,745** | **21,202** |

| | |
|---|---|
| **Discount Rate** | **10%** |
| *cumulated discount factor* | |
| Terminal cash flow | 21,202 |
| Terminal cost of capital | 10% |
| Terminal Value | 212,024 |
| Present Value (Terminal Value) | 81,745 |
| Present Value of next 10 years Cash Flows | 59,704 |
| **Total Present Value** | **141,448** |
| **Value of Operating assets =** | **141,448** |
| - Debt | 9,379 |
| - Minority Interests | 11,867 |
| + Cash | 3,824 |
| **Value of Equity** | **124,026** |
| Number of shares | 1,443 |
| **Estimated Value / share** | **85.9** |
| Price on market | 72.0 |
| ***Price as % Value*** | ***84%*** |




